Internal talk · 30 min

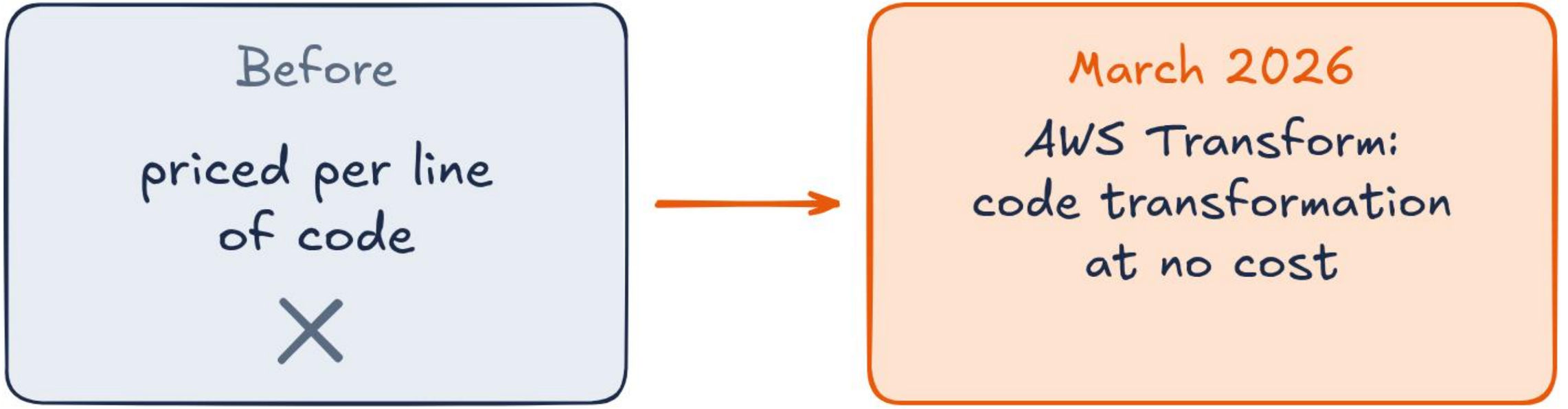
AI and legacy code

Translating is easy. Proving it is the job.



Translation is getting cheap

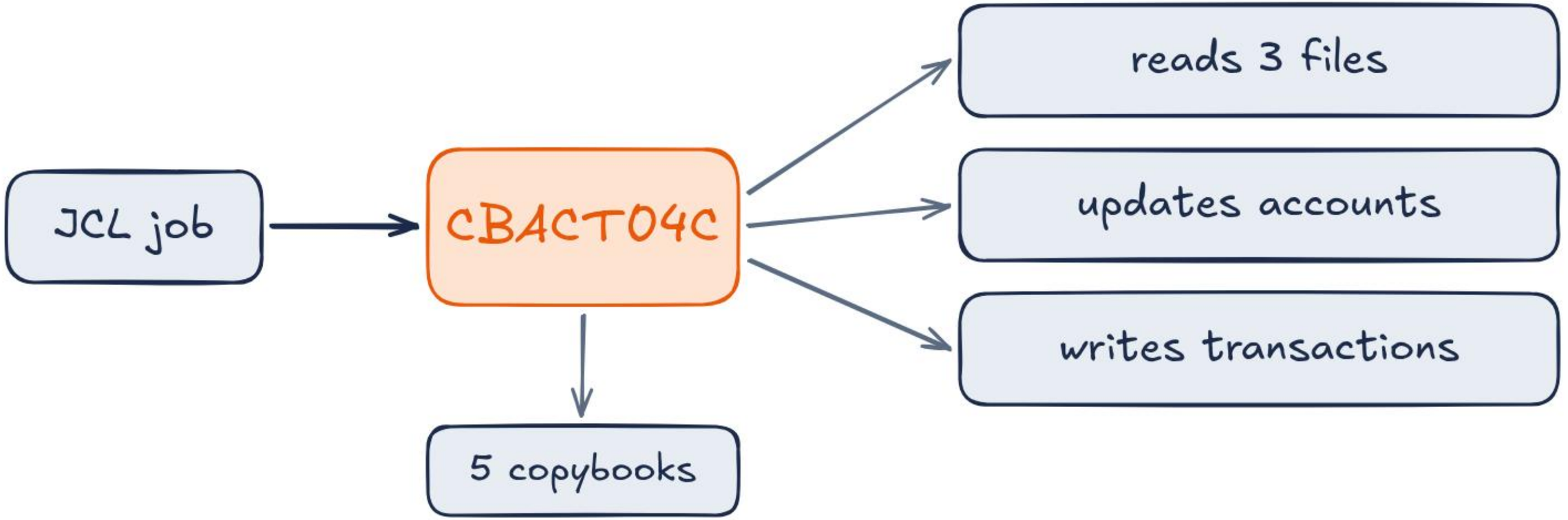
AWS announcement, 17 Mar 2026



So what are people paying for?

A program is not the system

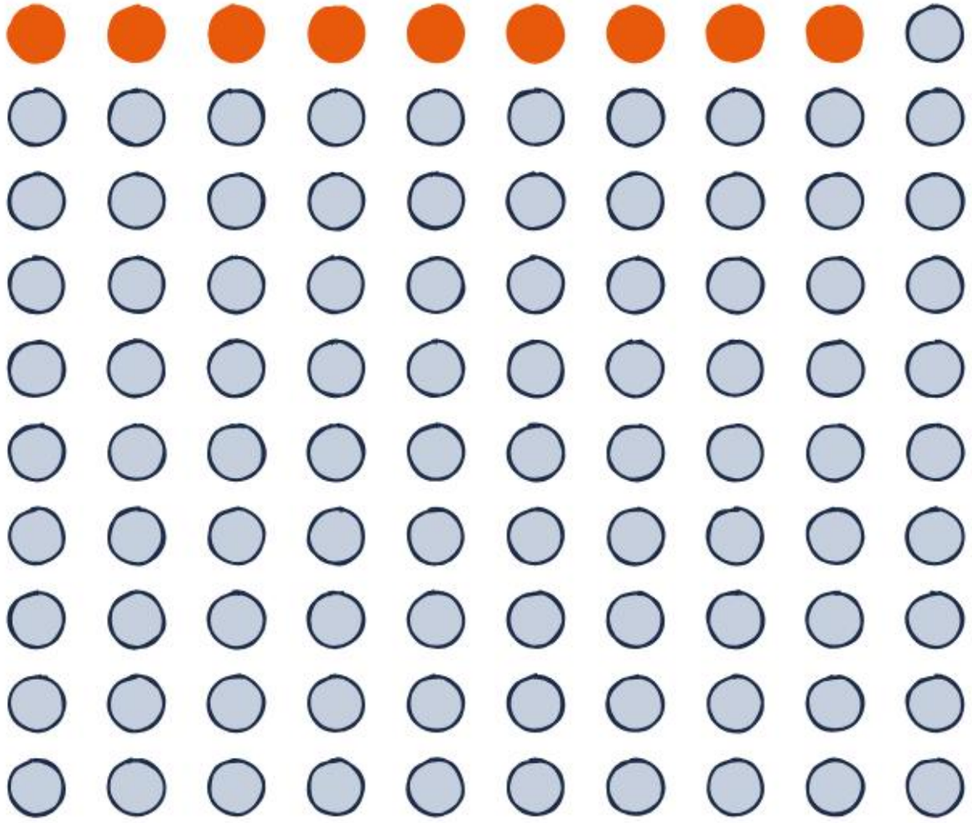
one interest job from AWS CardDemo, drawn by our dependency graph



+ an abend service

whole tree: 157 nodes, 385 edges

Few know the dependencies



9 %

fully understand their application dependencies

survey of 2,000 IT leaders, Oct 2026

Kyndryl, a services vendor

Two schools

Rule-based converters

provable,
but reads like COBOL in Java

LLM pipelines

idiomatic code,
but it must be proven



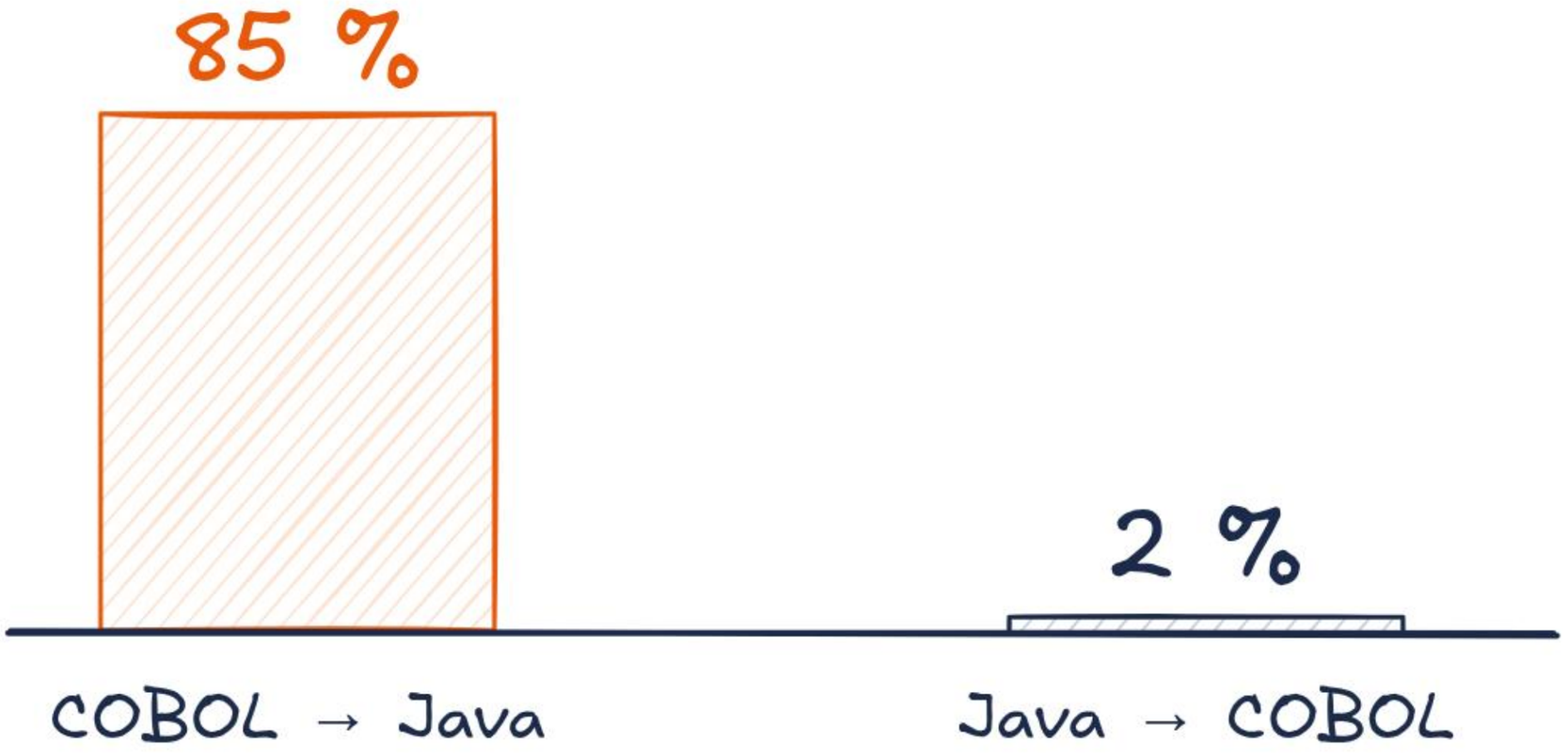


An oracle: the running system

everyone is converging here

Better readers than writers

tests passed by GPT-4o on small, single-file tasks



no copybooks, no job control, no other programs

COBOL-Coder paper, arXiv 2604.03986

Where it breaks

below the business rules

-5

How a minus sign is stored

12.49

Rounding or truncating



What a keyed file really does

999

Codes that schedulers watch

One missing word

```
COMPUTE WS-MONTHLY-INT =
  (TRAN-CAT-BAL * DIS-INT-RATE)
  / 1200

no ROUNDED ⇒ truncate
```

the old job cuts off the extra digits

it never rounds

our measurement

old COBOL
12.49

model, from COBOL only
12.50

One cent, on every transaction, every month

The forgotten last account

account 1	balance updated	✓
account 2	balance updated	✓
account 3	balance updated	✓
account 4	balance updated	✓
account 5	balance updated	✗

our measurement

Fix the bug?

9 of 25 scenarios still match

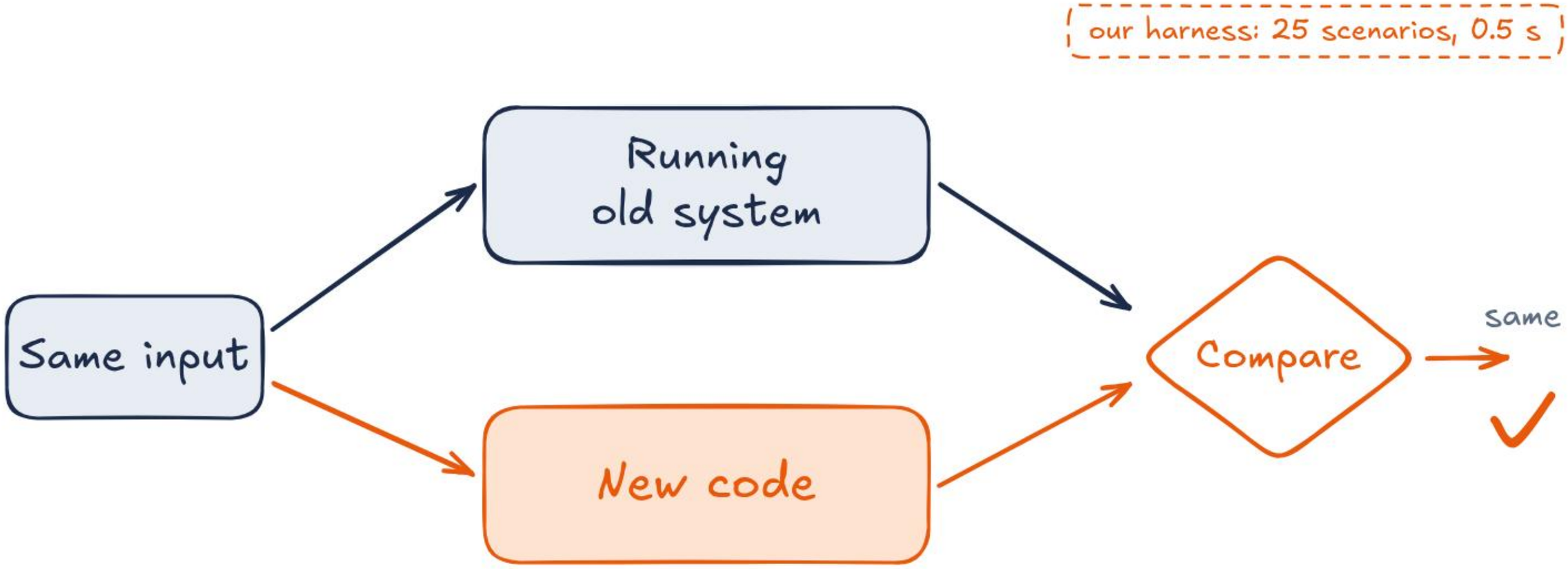
Copy it or fix it?

a business decision

the last one never gets its balance updated

found by running the job, not by reading it

The diamond



The running system is the spec

Live demo: the reference

```
$ python ex4_equivalence/diamond.py

carddemo-sample-data       ok    EQUIVALENT
edge-amounts               ok    EQUIVALENT
 ...
abend-unknown-account    abend 999   EQUIVALENT

25/25 scenarios equivalent in 0.5s
```

the real COBOL runs on a laptop

GnuCOBOL, unchanged source

shortened · recorded in a clean-clone run

workshop/ex4_equivalence/diamond.py

Live demo: a wrong candidate

```
$ python ex4_equivalence/diamond.py \
    --impl out/candidate_cobol.py

tx#1  amount  legacy=12.49  modern=12.50
tx#5  amount  legacy=0.99   modern=1.00
acct 9  curr_bal  legacy=560.00
                  modern=12500560.00
```

written by a free model from COBOL only

it rounded, and it posted the last account

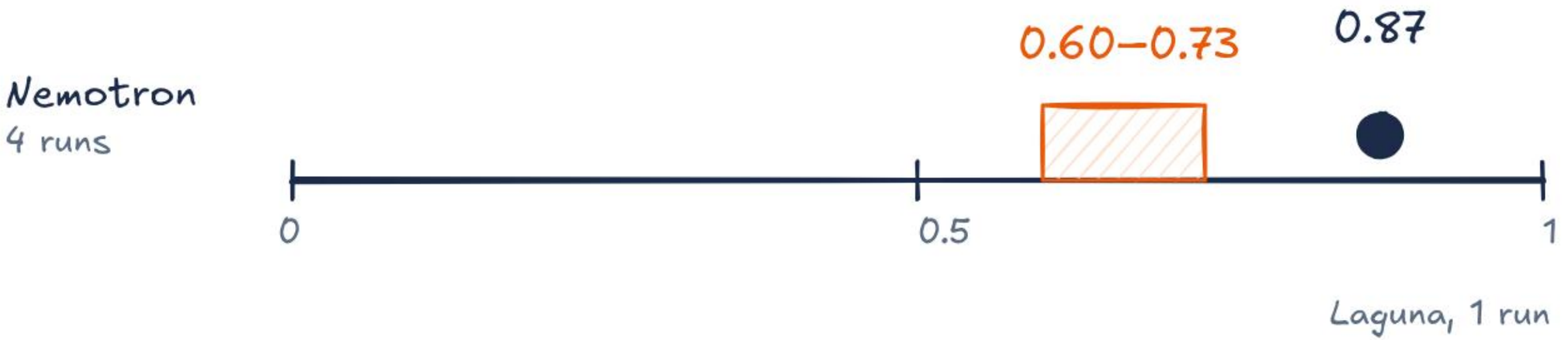
simplified from a recorded run

Looks like clean Python. Code review would miss it.

Reading rules with a model

how many of 15 hand-checked rules did the model find?

our measurement · free models



Always missed

- ✗ truncation, not rounding
- ✗ the last account is never posted

Cited, and still wrong

our measurement

Rule 17, lines 219–221

"At end of file, the program updates the last account's interest."

✓ right lines

✓ right symbols

✗ right meaning

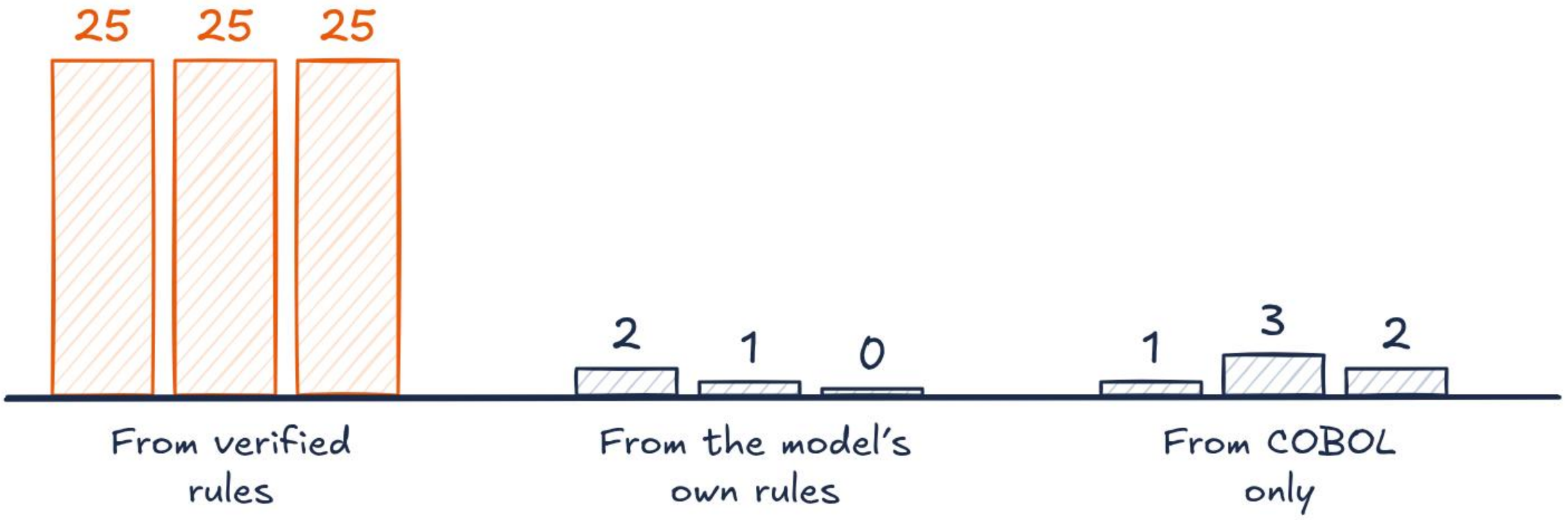
That branch never runs

A citation tells you where, not whether

Three ways to generate

scenarios that match the old system, out of 25

our measurement · free models



A later clean re-run of the first group scored 2 of 25: one wrong character.

Who sells what

from their docs · we did not run them

AWS Transform

code conversion at no cost

IBM Bob for Z

can run on your own servers

Google MAT + Dual Run

run both, compare outputs

Microsoft agents

open source, no built-in checks

Mechanical Orchard

verifies on real data flows

highlighted: the ones built around checking

Honest limits

One batch program

no online screens, no packed decimal

Free models

few runs, a small sample

Vendor tools not run

we read their documentation

Vendor numbers

are claims, prices mostly unpublished

Three takeaways

1. The running system is the spec
2. Trust the harness, not the model
3. Models draft, tests judge

Try it yourself

https://github.com/vusatui/rnd-research
→ legacy-modernization/

- Run the workshop
- Try the diamond
- Compare the vendors
- Read the sources



scan me

Questions?

AI and legacy code